we previously observed. In the case above, since x_0 sets a very high bar in terms of function values, neither x_1 nor x_2 constitute improvement, the kind of improvement we care about in optimization. In other words, a reasonable goal in optimization is to seek to *improve* from the best point that we have seen so far—as long as we improve from the best-seen point, we are making progress.

IMPORTANT **Definition of the incumbent**

The best-seen point, or the point that yields the highest function value that we have found so far, is often called the incumbent. This term denotes the fact that this point currently holds the highest value among all points queried during our search.

Given that we have a GP belief about the objective function, inspecting how likely or how much we can expect to improve from the best-seen point may be easily accomplished. Let's start from our running example of the Forrester function, where our current GP is shown in Figure 4.6.

Out of the two data points in our training set, *(x = 1, y = 1.6)* is the better one, as it has a higher function value. This is our current incumbent. Again, we care about improving from this 1.6 threshold. That is, we'd like to find data points that will yield function values higher than 1.6. Visually, we can imagine this improvement-based idea as cutting off the GP horizontally at the incumbent, as shown in Figure 4.8.

The portion highlighted darker corresponds to "improving from the incumbent" (at 1.6). Anything below this line will not constitute improvement, and therefore won't give us any additional optimization utility. Whether a point, for instance $x = 0$, will yield a higher function value is unknown, but we can still try to reason about the probability that it is true using what we know from the GP model.

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